- [CPSC 375: High-Performance Computing]

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Spring 2026

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Homework 31

NOTE: You are not required to hand in the following exercises, but you are strongly encouraged to complete them to strengthen your understanding of the concepts covered in class.

1. The scalability of parallel matrix multiplication is often limited by the ratio of computations per communication. In the block-striped algorithm, each process undergoes *p-1* block shifts. Using the complexity analysis from slide 16, derive why the block-striped computation-to-communication ratio of n/p is considered poor compared to Cannon's ratio of $n/\sqrt{p}$.

2. Both Cannon's and Fox's algorithms utilize a $\sqrt{p} \times \sqrt{p}$ checkerboard decomposition.
   A. In Cannon's algorithm, block A_{ij} is cycled left i positions and block B_{ij} is cycled up j positions during initial alignment. Trace the specific movement of blocks $A_{1,2}$ and $B_{1,2}$ in a 4×4 processor mesh during this phase.
   B. Compare the memory overhead of Fox's algorithm, which requires an extra buffer to store the broadcasted A block, against Cannon's algorithm, which is strictly in-place regarding communication buffers.
   C. Identify a hardware scenario where `MPI_Bcast` optimization might make Fox's algorithm faster than the manual shifts used in Cannon's.

3. SUMMA is designed to be highly scalable using a panel-based approach.
   A. Explain how rewriting the matrix product into a sum of outer products $C = \sum A_{,k} \times B_{k,}$ allows SUMMA to be more flexible than algorithms requiring a perfect square number of processors.
   B. In SUMMA, the block size b is chosen to balance communication latency and computational throughput. Explain how a very small b might increase latency overhead while a very large b might reduce the efficiency of pipelining.

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